Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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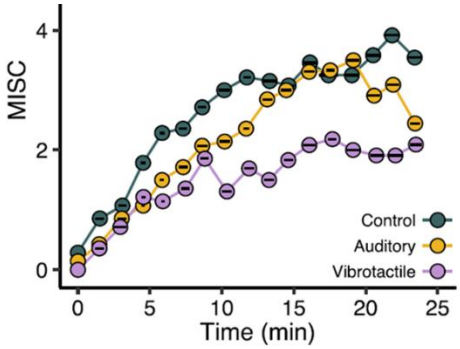
2. Technology Overview

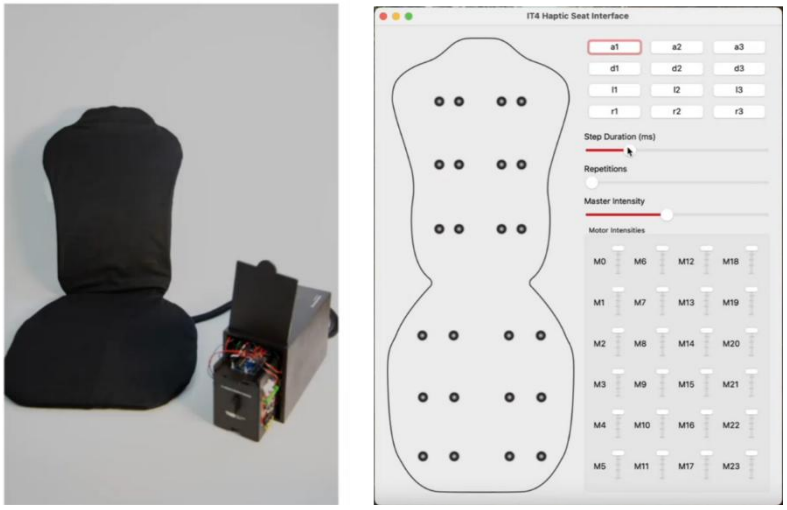
Title of Proposed Technology	**Vibro-tactile Motion Cueing Seat for Mitigating Motion Sickness in Autonomous Vehicles**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	Vehicle Intelligence Technologies	Automobile / Rail Vehicles	Safety Improvement Technologies
Concept of Proposed Technology	**Problem & significance** Motion sickness is a persistent barrier to passenger acceptance of automated vehicles (AVs). As automation increases, all occupants become passengers and are more likely to read or view screens, exacerbating sensory conflict (the mismatch between sensed and expected motion). Estimates suggest 30-65% of passengers experience carsickness in conventional conditions, with higher incidence expected in automated scenarios due to less involvement with driving tasks [1]. **Proposed solution** The proposed Vibro-tactile Anticipatory Motion Cueing Platform delivers vibro-tactile cues via the seat that precede and mirror vehicle manoeuvres (e.g., longitudinal & lateral accelerations, yaw rate), helping align predicted and actual sensory input for the occupants' perception system. Cues are low-intrusive (do not interfere with reading work, or audiovisual content), scalable in cost, and integrate into automotive seating. This is a solution direction that has already been researched by TNO, and proven to be a promising			

research direction for mitigating motion sickness. It also demonstrated that the effectiveness of this solution for the mitigation of motion sickness is highly dependent on the anticipation (cueing in advance of the motion itself), the vibratory patterns and intensity of the vibrations [2], [3].



Self-reported motion sickness severity (MISC) over time without anticipatory cues (green), with auditory cues (yellow) or with vibrotactile cues (purple). Adapted from Reuten et al. (2024).

In a recent project, TNO developed a highly customizable vibro-tactile cueing test bed that can be used for research in optimising the solution towards scale-up in the market. This will be used as the foundation of this project.



Vibrotactile motion cueing prototype (left) with software interface (right). The prototype contains 24 vibration motors, distributed across the seat pan and back rest.

In the next steps towards the market, additional research and development effort is required for the following activities:

1. Interface the vibro-tactile motion cueing platform with the automated driving system of the vehicle.
2. Optimising the following parameters:
    a. **Motion cueing patterns:** pattern, frequency content, actuator grouping.
    b. **Intensity:** based on road class, dynamics of the vehicle and passenger profile.
    c. **Anticipation** (ms-s scale): Tuned to account for vehicle dynamics and human response latency.
3. In-(automated) vehicle testing.
4. Productization.

This research project aims to take these steps towards getting the innovation to the market.

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Intended application areas

- L3 (or higher) automated and autonomous passenger cars and robotaxis (shuttles).
- Commercial vehicles/coaches (longer exposures).
- Adjacent markets**:** This also enables applications in the marine/rail/VR domains (provides additional future scope and possibility for broader valorisation of knowledge through methodology transfer).

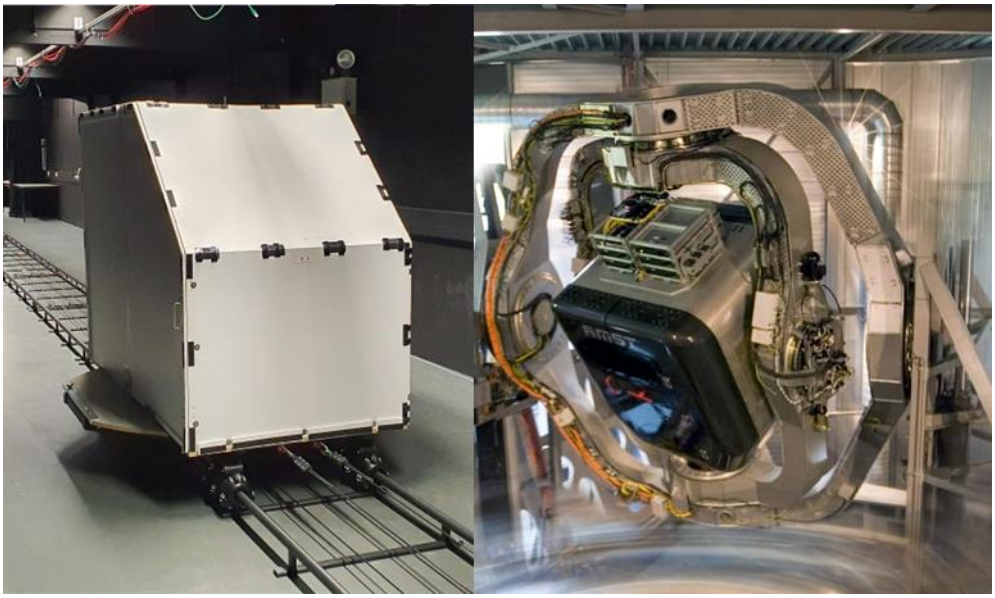
Rationale for Joint R&D Project

Research gap to be addressed

Optimize the triple design space for vibro-tactile cues—pattern × intensity × anticipation—and demonstrate on-road reduction in motion sickness incidence/severity under AV-like conditions.

Complementarity

- **TNO:**
  - Over 30 years of prior experience in motion sickness & human factors research, across the automotive, defense, space, cyber, maritime and other sectors.
  - TNO has specialized research facilities that can facilitate this research, including the Desdemona (6-DoF Motion Simulator), Limosine (40 m linear simulator), hexapod driving simulator, rotation chair, and TNO Carlabs (automated test vehicles).
- **Korean SME(s):**
  - Advanced seating/actuation hardware, embedded electronics, industrial design for integration and scale.



TNO motion simulators: Limosine linear motion simulator (left); Desdemona 6 DoF simulator (right).

Why international collaboration

Mitigation of motion sickness requires interdisciplinary co-development (vehicle dynamics + human factors + physiology) and multi-demographic validation.

AV supply chains are inherently global. The opportunity to co-design the solution with Korean SMEs (working on seating, actuators, control hardware) accelerates the possibility to scale-up the innovation from TNO, towards application in the automotive industry.

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	Outcome Optimized, seat-integrated platform with a quantified benefit in the mitigation of motion sickness , and validation data suitable for productization steps in Korea/EU.
Key Words	1. Mitigation of motion sickness in autonomous vehicles. 2. Next generation automotive seating. 3. Anticipatory vibro-tactile motion cueing. 4. Increased occupant trust, acceptance, comfort and perceived safety in AVs.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	• **Integration.** Demonstrate in-vehicle operation, interfaced with an automated test platform. • **Develop the algorithm.** Develop a real-time cueing engine that converts vehicle state (preview + current dynamics) into anticipatory vibro-tactile commands for seat actuators. • **Research on effectiveness.** Quantify reduction in motion sickness incidence and severity using validated scales (e.g., MISC, MSAQ) under scripted AV maneuvers and naturalistic driving. • **Productization.** Build a prototype that is closer to production (automotive-grade actuators, silent operation, thermal-safe, low power) with an interface to the vehicle's motion planning platform and EMC pre-compliance plan. • **Path to scale-up.** Produce Design & Integration Guidelines, test protocols, and data packages for SME adoption and scale-up.			
R&D Target Specifications	Comfort impact • ≥ 40% reduction in motion sickness incidence vs. baseline (no cues) in AV-representative scenarios (primary endpoint). • ≥ 1 point increase in perceived occupant comfort (measured on a 5-point scale) under realistic automated driving conditions (secondary endpoint). Cue Engine Performance • Anticipation timing control: from 0s up to 2s before the maneouver. Actuation and integration: • ≤ 30 dBA @ 0.5 m in cabin background (per ISO-based cabin noise checks). • Power: ≤ 15 W average per seat during heavy maneuvers; passive standby ≤ 0.5 W. • Thermal: surface temperature increase ≤ 5 °C above ambient during prolonged use. Robustness & Acceptance • Intention to buy > 60% measured through a subjective measure. • No adverse events related to actuation (safety endpoint).			

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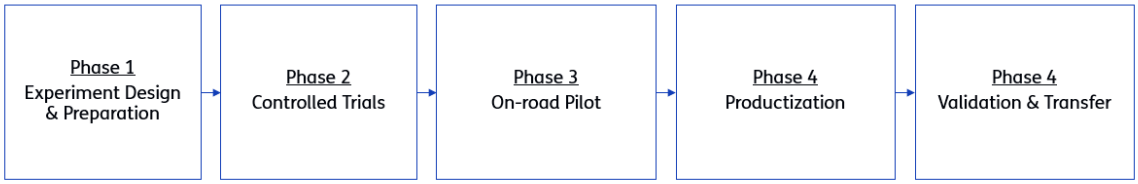
R&D Plan

TNO Proposal: Vibro-tactile Motion Cueing Seat for Mitigating Motion Sickness in Autonomous Vehicles



Phase 1: Experiment Design & Preparation (Months 0–9, Lead: TNO)

- Literature update (post-2020); define baseline scenarios (curvy roads, stop-and-go, lane changes, roundabouts).
- Human study design: sample size planning, randomization, counterbalancing, inclusion/exclusion, consent materials.
- Cue space definition: manoeuver description, pattern library (actuator groups, triggers etc.), intensity curves, anticipation windows tied to control/preview (map → maneuver → cue).
- Pilot experiment on a static simulator setup to specify the parameter space of interest for subsequent phases.
- Testbed: program simulator &/or on-vehicle pipelines; interfacing between the motion planner and the motion cueing platform.
- Ethics: Internal Review Board dossier; risk mitigation (nausea management, stop criteria).

Milestone M1: Approved protocol & finalized cue design matrix; simulator & vehicle pipelines validated.

Phase 2: Controlled Trials (Months 9–12, Lead: TNO)

- Simulator study (n ≈ 40–60): repeated-measures across cue vs. baseline conditions; collect MISC/MSAQ, exposure time, task performance (reading), physio (optional).
- Analysis: primary/secondary endpoints, subgroup explorations (susceptibility levels).

Milestone M2: Statistically significant effects identified; short-list cue configurations.

Phase 3: On-road Pilot (Months 12–15, Lead: TNO)

- Supervised automated test vehicle running on a fixed test route with a range of different manoeuvres as they occur in normal driving (changes in speed, braking, cornering, lane changes).
- Data: vehicle dynamics, cue logs, participant reports.

Milestone M3: On-road feasibility & effect size estimates; updated safety & HMI guidelines.

Phase 4: Productization (Months 12-20, Lead: SME)

- Seat integration: actuator placement, mechanical isolation, cable routing, silent operation.
- Electronics: embedded controller, diagnostics, failsafe.
- Firmware: real-time cue scheduler, calibration routines, logging.
- Bench tests: durability (e.g., >1,000 cycles), thermal soak, EMC pre-checks.

Milestone M4: Integrated TRL5 prototype seat; bench validation completed.

Phase 5: Validation & Transfer (Months 20–24, Lead: SME)

- On-road validation with optimized cues; A/B comparisons vs. baseline.
- Documentation: Integration Guide, Test Protocol Pack, Safety Case summary, UX/HMI guide.

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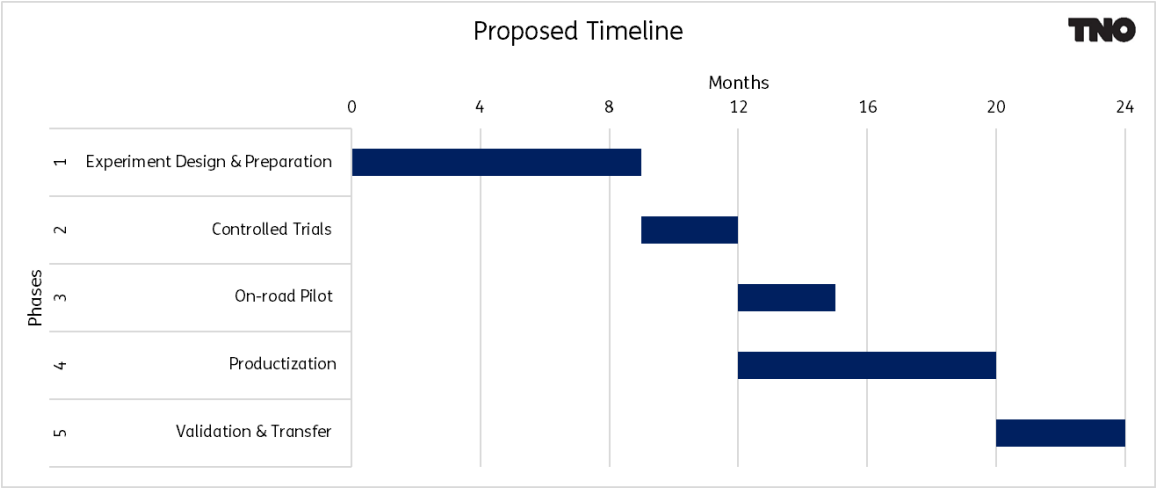
Milestone M5: TRL6 demonstration in relevant environment.



Evaluation & Measurement Plan

- **Primary metric:** Motion sickness incidence (% of participants reaching MISC ≥ threshold or discontinuing due to nausea) per condition.
- **Secondary metrics:** Symptom severity (MISC trajectory, SSQ post-exposure), time-to-onset, self-reported comfort/acceptance.
- **Statistical approach:** within-subject analyses (e.g., mixed-effects models for condition × scenario), confidence intervals; correction for multiple comparisons where relevant.
- **Success criteria:** meet/exceed target reductions; demonstrate no negative trade-offs (e.g., distraction, annoyance).

Data, Ethics & Safety

- **Ethics/IRB:** consented adult volunteers; stopping rules for emesis.
- **Privacy/GDPR:** pseudonymized data, role-based access, clear retention policy, deletion upon request.
- **On-vehicle safety:** Constant supervision; manual override switches.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	○ Lead research: experiment design, IRB, execution, analysis. ○ Interfacing with TNO automated driving platform: develop the interface with the TNO Carlab test platform. ○ Algorithm development: anticipation timing, pattern selection, intensity adaptation. ○ Validation: driving simulator and public-road pilots; documentation and transfer materials. Provide access to relevant facilities: Desdemona, Limosine, hexapod & driving simulators, automated vehicle platforms.
Expected R&R of Korean SME Partner	○ Productization: manufacturable actuator selection, packaging, and cost targets. ○ Interface specification and pathways definition. ○ Testing and validation of the test platform in Korea. ○ Commercialization planning: roadmap to OEM/Tier-1 integration.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical outcomes** ○ A seat-integrated anticipatory vibro-tactile cueing system reaching TRL6 in AV-relevant use. ○ Cueing engine with tunable anticipation, pattern, and intensity; interface & diagnostics ready for Tier-1 integration. ○ Validated protocols and datasets demonstrating reduction in motion sickness in representative scenarios. **Industrial & economic impact** ○ Differentiated comfort feature for Korean automotive suppliers and OEMs. ○ Supports AV adoption by addressing a crucial discomfort barrier. ○ Improves trust and perceived safety for occupants of an automated vehicle. **Cross sector valorisation potential** ○ Potential to scale applications to other sectors, such as maritime, defense etc.

6. References

[1] C. Diels and J. E. Bos, “Self-driving carsickness,” *Applied Ergonomics*, vol. 53, pp. 374–382, Mar. 2016, doi: 10.1016/j.apergo.2015.09.009.

[2] A. J. C. Reuten, J. B. J. Smeets, J. Rausch, M. H. Martens, E. A. Schmidt, and J. E. Bos, “The (in)effectiveness of anticipatory vibrotactile cues in mitigating motion sickness,” *Exp Brain Res*, vol. 241, pp. 1251–1261, Mar. 2023, doi: 10.1007/s00221-023-06596-8.

[3] A. J. C. Reuten, I. Yunus, J. E. Bos, M. H. Martens, and J. B. J. Smeets, “Anticipatory cues can mitigate car sickness on the road,” *Transportation Research Part F: Traffic Psychology and Behaviour*, vol. 105, pp. 196–205, Aug. 2024, doi: 10.1016/j.trf.2024.07.006.